

3 In an investigation, the water potential of a potato was compared with that of a yam.

(a) State what is meant by the term **water potential**.

(2)

(b) In the investigation, six cylinders were cut from each of a single yam and a single potato.

In each case, one cylinder was placed in distilled water and the others were placed in solutions of different concentrations of sucrose (mol dm^{-3}). The mass of each cylinder was measured at the start and after one hour.

(i) Describe how the mass could be measured in a valid and accurate way.

(2)

Valid

Accurate

(ii) State the independent variable in this investigation.

(1)



- (iii) In this investigation, the tubes containing the cylinders and the solutions were kept at the same temperature.

Explain the effect on this investigation if the yam cylinders had been kept in solutions at 20°C and the potato cylinders in solutions at 25°C.

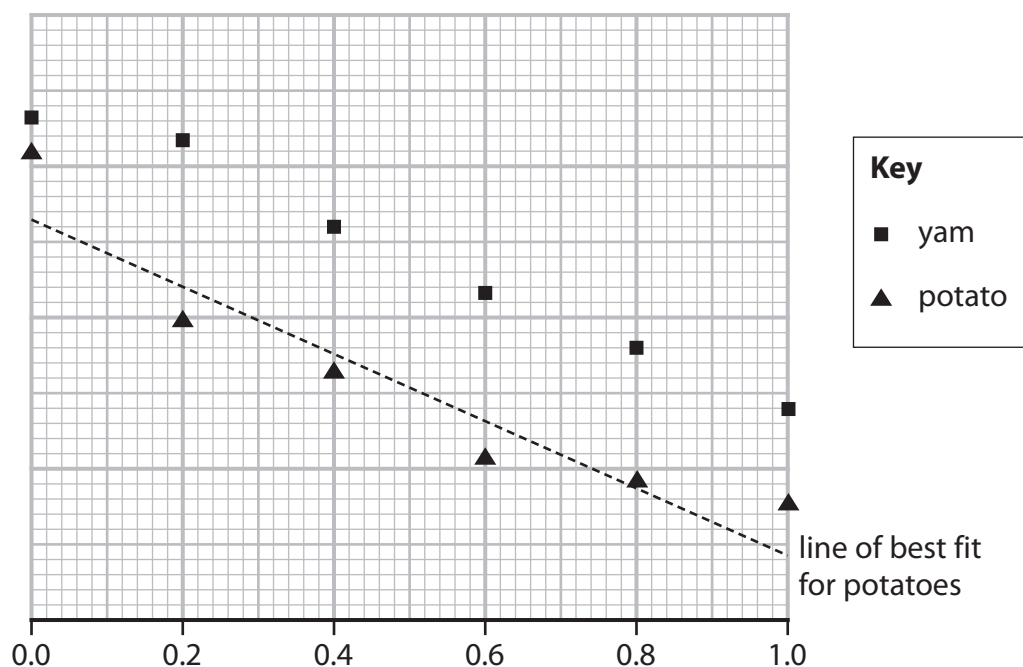
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- (c) The graph shows the results obtained in this investigation.



- (i) Complete the graph by adding a y-axis scale, y- and x-axes labels and a line of best fit for the yam.

The highest percentage change for the yam is 26% and the lowest percentage change for the yam is -12%.

(3)

- (ii) The equation for a straight line is $y = mx + c$.

Calculate m for the line of best fit for the potato data and write down the equation for this line.

(3)

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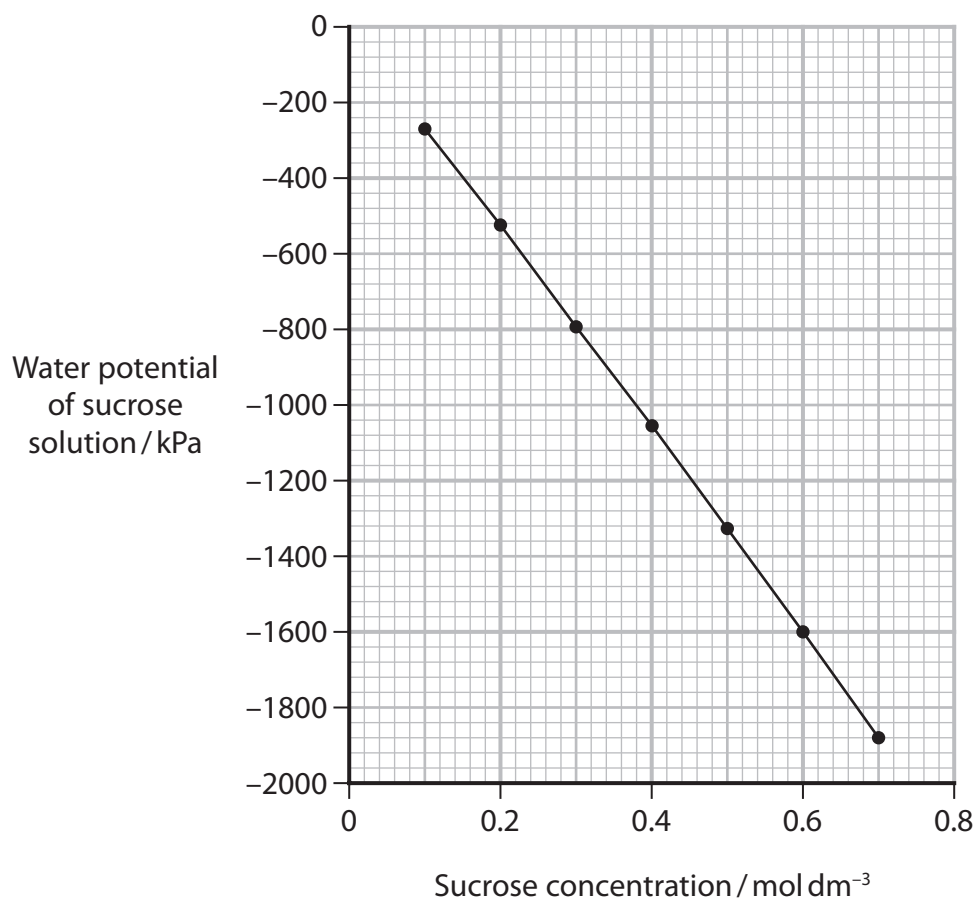
(iii) Different concentrations of sucrose solutions have different water potentials.

The graph shows the water potentials for sucrose solutions of different concentrations.

Determine the water potential for the potato cylinders.

Use your completed graph and this graph to determine the water potential.

(2)



Value read from completed graph

Water potential of the potato kPa



(iv) The water potential for the yam was -1890 kPa .

Suggest why there is a difference between the water potential of the yam and the water potential of the potato.

(1)

(Total for Question 3 = 17 marks)

TOTAL FOR PAPER = 50 MARKS

